

DESIGN REVIEW / 2026-09-26

Smaller FPGA PCB Components and every pin

XC7A100T, CSG324 / 15 × 15 mm package

Result: a 33 × 36 mm native KiCad placement, retaining all 128 components and both 60-contact mezzanine connectors. This removes 162 mm² (12%) from the earlier 37.5 × 36 mm study, and 252 mm² (17.5%) from the original 40 × 36 mm outline.

Electrical status: unfinished. This revision has no tracks, vias or copper zones. It preserves 47 assigned nets and has 383 unconnected items in KiCad. All 120 mezzanine signal contacts are still unassigned in the PCB. The proposed allocation of 117 ASIC signals plus three spare contacts is documented separately; it has not been wired to FPGA balls.

The report explains the reason for every retained component, lists every electrical endpoint, and identifies which connections exist only as CAD assignments. It does not certify that these values, the power system or the complete board are ready to manufacture.

Revision	Outline / area	Meaning
Original study	40 × 36 mm / 1,440 mm ²	Baseline connector-inclusive placement.
Earlier compact	37.5 × 36 mm / 1,350 mm ²	Preserved for comparison; 6.25% below baseline.
This revision	33 × 36 mm / 1,188 mm ²	All 128 parts retained; 39 front and 89 back.
Native DRC	0 physical violations	383 unconnected items; no integrated-schematic parity check.

How to use the report. Read the design overview first. Appendix A explains every component; B lists all 324 FPGA balls; C lists the cable and mezzanine contacts; D lists the converter, flash and oscillator pins; E expands every assigned net into its complete list of connected endpoints. A matching CSV set preserves exact raw KiCad names and every physical pad.

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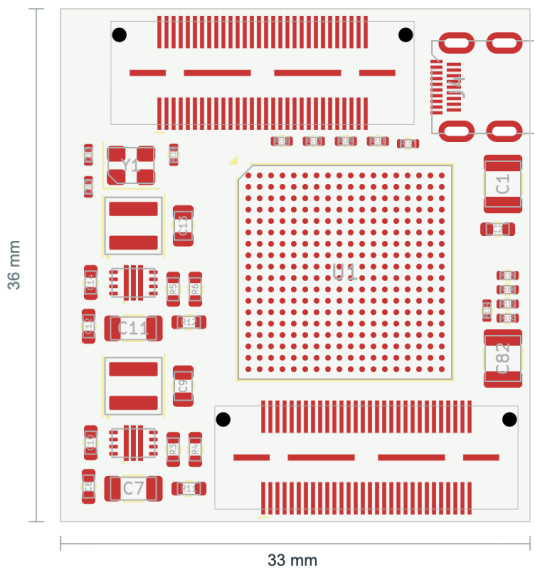
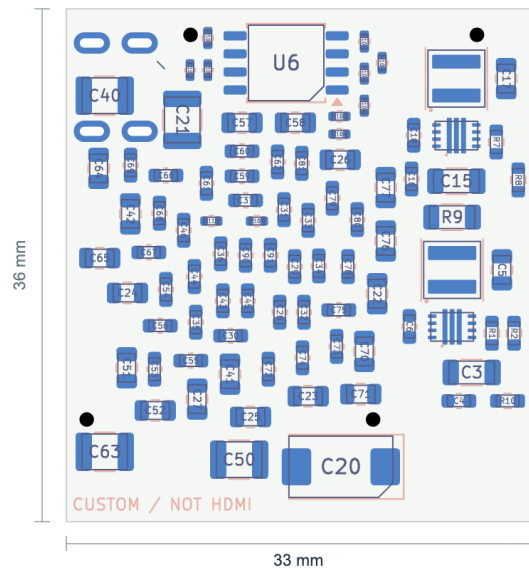
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1 Smaller layout and the remaining empty space

33 × 36 mm · 12% less board area · all 128 parts retained

Compared with the 37.5 × 36 mm study; 17.5% smaller than the original 40 × 36 mm connector-fit study.

a Front · 39 parts

**b Back (mirrored) · 89 parts**

Unrouted placement study · native DRC: 0 physical violations / 383 unconnected items · not for fabrication

J4 body projects 0.65 mm beyond the board; insertion and mating-board clearance remain unverified.

Figure 1. Native 33 × 36 mm placement, front and mirrored back. Only component geometry is shown; no connecting copper is present. The PCB rectangle is not the complete plug or mating-board envelope.

The reduction comes from repacking both faces and moving J5, J4, the back-side power groups, the clock group and selected support parts inward. U1 and J6 stay fixed. The existing converter groups retain their internal geometry. No decoupling capacitor, connector, flash, oscillator, strap or regulator was removed to obtain the smaller outline. There are no added debug headers, user LEDs or standalone test points in this candidate.

The front openings in the earlier core-only image are occupied by the two required mezzanine connectors in the complete placement study. The long back-side opening is mirrored relative to the front and can overlap a regulator or its required routing area on the opposite face. Empty drawing space therefore does not directly equal removable board area.

This is a demonstrated smaller placement, not proof of the smallest functional board. BGA escape, the power return paths, connector mating height, cable strain, protection components and thermal copper may require space that is currently blank. Assembly tolerances must be checked in addition to the saved CAD rules. **Measured limits.** The smallest same-face courtyard-box gap is about 0.010 mm; the smallest fabrication-body-box gap is 0.360 mm. The nearest opposite-face courtyard-to-through-pad-box gap is 0.090 mm. The minimum copper-pad edge margin is 0.500 mm. These are native-geometry measurements, not approved assembly allowances.

J4's drawn body extends 0.65 mm beyond the PCB edge, giving a board-plus-drawn-body envelope of approximately **33.65 × 36 mm**; its courtyard extends 1.125 mm. The full mated assembly and plug envelope remain unverified. The earlier 37.5 mm and intermediate 34 mm candidates are preserved if routing or assembly needs more space.

2 How the intended circuit works

The following paths describe *logical net assignments*. Every arrow still requires real copper in this placement. Parts and rails belong to the inherited 3.3 V boot/configuration version. The separate 1.8 V configuration / 2.5 V link-bank proposal is not integrated here.

2.1 Power and sequencing

J4.19 is assigned to LINK_12V. It feeds the input capacitors and all four converters. This is a proposed custom power connection, not proof that an XEM8310 carrier or an ordinary micro-HDMI cable can safely supply it.

Path	Assigned circuit	Why it exists
Core	U2.2 → L1 → VCORE_REG_1V025 → R9 → VCCINT_1V0	Powers FPGA logic and block RAM. R9 is a 15 milliohm series element.
Auxiliary	U3.2 → L2 → VCCAUX_1V8	Powers FPGA auxiliary and analog supply domains.
Boot / config	U4.2 → L3 → VCC_CFG_3V3	Supplies configuration I/O, flash, oscillator and pull-ups.
ASIC-facing I/O	U5.2 → L4 → VCC_ASIC_1V5	Supplies selected FPGA I/O banks; external ASIC power delivery is not established.
Enable chain	LINK_12V enables U2; PG_CORE enables U3; PG_AUX enables U4 and U5.	Power-good sequencing intent. Startup and discharge behaviour still require validation.

The four TPS62135 converters use a local inductor, input and output capacitors, feedback divider and soft-start capacitor. Pin 11 is **VSEL**, tied to ground; it must not be mistaken for a second ground pin. Pin 3 is the ground pin. Pin 4 (FB2) is intentionally left open in this configuration. The pin roles were checked against TI's datasheet [1].

U2 senses the output *before* R9. The nominal core voltage after R9 changes with current: a 1 A load produces about 15 mV drop and 15 mW dissipation in a 15 milliohm element. That arithmetic is not a load or thermal validation. FPGA activity, converter limits, capacitor bias loss and the actual supply tolerance still need analysis.

2.2 Configuration, clock and programming

U6 is the proposed 3.3 V configuration flash. U1.E9 drives its clock through R103; DQ0–DQ3 use R104–R107; U1.L13 controls chip select. The complete endpoint lists are in Appendices B–E. U6.7 is RESET/SIO3 for the stated Macronix part, not an assumed HOLD input [4].

R111–R113 encode M[2:0] = 001; R108–R110 pull up PROGRAM_B, INIT_B and DONE. R114 pulls PUDC_B high. These inherited assignments document the master-SPI boot intent; compatible firmware, boot settings, reset recovery and a power-up test are not yet demonstrated [2].

Y1 provides a separate 32 MHz user clock. Y1.3 passes through R119 to U1.P17. Y1.1 is tied high to enable the oscillator; pins 2 and 4 are ground and supply [5]. JTAG uses J4.2/15/17/18 and dedicated FPGA pins, with R118 in the TDO path. J4.1 is a 3.3 V target-reference assignment, not an independently qualified power output.

3 Pin budget, terminology and completeness

The team correction is **117 ASIC-related signals**: 88 ASIC output/interface signals, 12 SPI data/control streams, nine shared controls, four board clocks and four additional SPI clocks. This is an interface budget, not a complete FPGA package pin count.

Group	Count	Interpretation
ASIC output/interface	88	Eight groups of 11: 64 DATA1–8 lines plus eight each of CLK32MHz_Out, READ and SYNC.
SPI data	12	Four stimulation SPI L, four stimulation SPI R, two non-stimulation SPI L and two non-stimulation SPI R.
Shared controls	9	CHIP_RESET, AC_IN, IMP_TST, FE_RESET, SPI_LATCH, STIM_CLK, S_TIM_START, STIM_EN, STIM_CHB
Board clocks	4	One board clock per board in the supplied requirement.
SPI clocks	4	The four SPI_CLK connections added in the team correction.
Total	117	Proposed across 120 mezzanine contacts, with three contacts spare.

AC_IN and IMP_TST remain electrically unresolved: their names and tutorial diagrams alone do not establish safe direct FPGA voltage limits or whether conditioning is required. The full ASIC pad specification and timing contract are needed before allocating FPGA balls. The XEM8310 is an FPGA module used as the downstream endpoint; the required carrier, protocol, clocking and supply connection remain separate design work.

Assigned means that KiCad places endpoints on the same named logical net. **Unrouted** means no copper path has been built in this revision. **Unassigned** means no functional net has been allocated. **Intentional NC** marks a pin deliberately left open in the inherited core design; it is not a spare signal.

Coverage	Count	Meaning
Physical components	128	82 capacitors, 32 resistors, four inductors, six ICs, three connectors and one oscillator.
Unique electrical endpoints	758	421 assigned, 331 unassigned, six intentional NC. Every endpoint is listed in the appendices.
Numbered physical pads	767	Includes four shell pads sharing J4.SH, four ground lands sharing J5.G and four sharing J6.G.
Unnumbered records	40	36 solder-paste-only apertures and four non-plated connector locating holes; these are not extra electrical pins.
All physical pad records	807	767 numbered plus 40 unnumbered; preserved from the earlier placement.
Functional nets	47	KiCad placeholder unconnected names are excluded. The full peer list of each real net appears in Appendix E.
Unassigned breakdown	331	203 FPGA user-I/O balls, eight reserved cable contacts, 120 mezzanine signal contacts.
Intentional NC	6	U1.L9 / L10 (DXN / DXP) and U2–U5 pin 4 (FB2).

4 What must be resolved before release

The 33×36 mm result improves placement. It does not close the earlier system uncertainties. The following issues directly affect whether this size can become a functioning board.

Open item	Required next evidence / proposed owner
ASIC electrical contract	ASIC design owner / Gerald: confirm all pad types, levels, drive strengths, reset states and setup/hold limits, especially AC_IN and IMP_TST.
FPGA assignment and bank rails	FPGA engineering: select all 117 FPGA balls and compatible banks/standards, then generate the schematic and XDC from one checked pin table.
Downstream link and power	FPGA + carrier engineering: define a realizable XEM8310 carrier connection, data protocol, clocking, cable pinout, current limit, length and protection.
One integrated circuit	PCB engineering: reconcile this 3.3 V core-derived layout with the separate rail revision, including C92/R122 and any selected replacement flash/oscillator.
Routing and stackup	PCB engineering + fabricator: route all power, ground, boot, clock, JTAG, ASIC and link nets; agree BGA escape rules, stackup and impedance.
Mechanical assembly	PCB engineering + assembler: approve the connector body overhang, mating-board offsets, height, retention, courtyard tolerances and dual-side assembly access.
Part values and procurement	PCB engineering: finish all TBD order codes, check footprint/part matches and qualify capacitor effective values and the required decoupling count.
Measured acceptance	FPGA + lab team: cold boot, brownout/recovery, all-channel capture, sustained transfer, error rate, power and thermal tests on actual hardware.

The current complete-board design has no manufacturing release. A DRC pass does not demonstrate signal integrity, power integrity, thermal margin, assembler acceptance or working firmware. It also does not check the 117 signals that have no assigned PCB net yet.

Further reduction path. The next useful size decision is an integrated routing trial with the required protection and bank-voltage choices. A changed connector orientation or a smaller power architecture may reduce size further, but cannot be approved by deleting components or trimming blank drawing space alone.

A Every component: purpose and connections

All 128 physical components appear below. Values and order codes describe the saved candidate; TBD is preserved wherever selection is incomplete. The table is a design ledger, not an approved purchase BOM. For two-terminal components, both pins are listed here. Larger devices refer to their complete pin tables later in this report.

Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C1 47u 25V X5R Front	Cable-input bulk charge reservoir between LINK_12V and ground. Does not replace input protection. Order: MSAST32MAB5476MPNDT1	1: LINK_12V 2: GND
C2 100n 100V X7R Front	Cable-input local high-frequency bypass between LINK_12V and ground. Does not replace input protection. Order: GRM188R72A104KA35D	1: LINK_12V 2: GND
C3 22u 25V X5R Back	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	1: LINK_12V 2: GND
C4 100n 100V X7R Back	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	1: LINK_12V 2: GND
C5 22u 16V X5R Back	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	1: VCORE_REG_1V025 2: GND
C6 10n 50V X7R Back	Soft-start capacitor from SS_CORE to ground; sets the converter reference ramp. Order: TBD - manufacturer/order code not selected	1: SS_CORE 2: GND
C7 22u 25V X5R Front	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	1: LINK_12V 2: GND
C8 100n 100V X7R Front	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	1: LINK_12V 2: GND
C9 22u 16V X5R Front	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	1: VCCAUX_1V8 2: GND
C10 10n 50V X7R Front	Soft-start capacitor from SS_AUX to ground; sets the converter reference ramp. Order: TBD - manufacturer/order code not selected	1: SS_AUX 2: GND
C11 22u 25V X5R Front	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	1: LINK_12V 2: GND
C12 100n 100V X7R Front	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	1: LINK_12V 2: GND
C13 22u 16V X5R Front	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	1: VCC_CFG_3V3 2: GND
C14 10n 50V X7R Front	Soft-start capacitor from SS_CFG to ground; sets the converter reference ramp. Order: TBD - manufacturer/order code not selected	1: SS_CFG 2: GND
C15 22u 25V X5R Back	Local converter input reservoir on LINK_12V. Nominal 22 uF / 25 V; effective capacitance at bias and placement need qualification. Order: MSAST31LBB5226MTNA01	1: LINK_12V 2: GND

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C16 100n 100V X7R Back	Small local converter input bypass from LINK_12V to ground. Order: GRM188R72A104KA35D	1: LINK_12V 2: GND
C17 22u 16V X5R Back	Local converter output capacitor and voltage-sense node; keep close to inductor/ground return. Order: EMK212BBJ226MG-T	1: VCC_ASIC_1V5 2: GND
C18 10n 50V X7R Back	Soft-start capacitor from SS_ASIC to ground; sets the converter reference ramp. Order: TBD - manufacturer/order code not selected	1: SS_ASIC 2: GND
C20 330u 2.5V polymer ESR25m Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: T520V337M2R5ATE025	1: VCCINT_1V0 2: GND
C21 100u 6.3V X5R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER60J107ME20L	1: VCCINT_1V0 2: GND
C22 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C23 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C24 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C25 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C26 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C27 4.7u 10V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCINT_1V0 2: GND
C28 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C29 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C30 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C31 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C32 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C33 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C34 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C35 470n 6.3V X7R Back	Distributed logic-core decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C36 470n 6.3V X7R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C37 470n 6.3V X7R Back	Distributed block-RAM decoupling on VCCINT_1V0. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCINT_1V0 2: GND
C40 47u 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	1: VCCAUX_1V8 2: GND
C41 4.7u 10V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCAUX_1V8 2: GND
C42 4.7u 10V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCCAUX_1V8 2: GND
C43 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCAUX_1V8 2: GND
C44 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCAUX_1V8 2: GND
C45 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCAUX_1V8 2: GND
C46 470n 6.3V X7R Back	Distributed auxiliary decoupling on VCCAUX_1V8. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCCAUX_1V8 2: GND
C50 47u 6.3V X7R Back	Distributed configuration-bank decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	1: VCC_CFG_3V3 2: GND

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C51 4.7u 10V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_CFG_3V3 2: GND
C52 4.7u 10V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_CFG_3V3 2: GND
C53 470n 6.3V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C54 470n 6.3V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C55 470n 6.3V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C56 470n 6.3V X7R Back	Distributed I/O bank 14 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C57 4.7u 10V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_CFG_3V3 2: GND
C58 4.7u 10V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_CFG_3V3 2: GND
C59 470n 6.3V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C60 470n 6.3V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C61 470n 6.3V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C62 470n 6.3V X7R Back	Distributed I/O bank 16 decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_CFG_3V3 2: GND
C63 47u 6.3V X7R Back	Distributed configuration / I/O bulk decoupling on VCC_CFG_3V3. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	1: VCC_CFG_3V3 2: GND
C64 4.7u 10V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C65 4.7u 10V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND
C66 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C67 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C68 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C69 470n 6.3V X7R Back	Distributed I/O bank 15 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C70 4.7u 10V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND
C71 4.7u 10V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND
C72 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C73 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C74 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C75 470n 6.3V X7R Back	Distributed I/O bank 34 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C76 4.7u 10V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND
C77 4.7u 10V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM21BR71A475KA73L	1: VCC_ASIC_1V5 2: GND
C78 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
C79 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C80 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C81 470n 6.3V X7R Back	Distributed I/O bank 35 decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM188R70J474KA01D	1: VCC_ASIC_1V5 2: GND
C82 47u 6.3V X7R Front	Distributed ASIC-facing I/O bulk decoupling on VCC_ASIC_1V5. This inherited value/count is a design candidate, not a demonstrated minimum. Order: GRM32ER70J476ME20L	1: VCC_ASIC_1V5 2: GND
C90 100n 16V X7R Back	Bypass for the FPGA analog/XADC supply domain, tied to the auxiliary rail in this design. Order: TBD - manufacturer/order code not selected	1: VCCAUX_1V8 2: GND
C91 1u 10V X7R Back	Bypass for the FPGA analog/XADC supply domain, tied to the auxiliary rail in this design. Order: TBD - manufacturer/order code not selected	1: VCCAUX_1V8 2: GND
C100 100n Back	Local flash supply decoupling from 3.3 V to ground. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: GND
C101 1u Back	Local flash supply decoupling from 3.3 V to ground. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: GND
C102 10n Front	Local oscillator supply decoupling from 3.3 V to ground. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: GND
C103 100n Front	Local oscillator supply decoupling from 3.3 V to ground. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: GND
J4 CUSTOM POWER + JTAG / DATA TBD Front	Single custom cable port for proposed 12 V input, JTAG and four reserved data pairs. It is not wired as ordinary HDMI. Order: Molex46765 family; exact order code pending	All cable contacts: Appendix C.
J5 QSH-030-01-L-D-A Front	60-contact routing-board mezzanine plus four ground-blade lands and two locating holes. Proposed ASIC contact allocation is separate from CAD. Order: QSH-030-01-L-D-A	All mezzanine contacts: Appendix C.
J6 QSH-030-01-L-D-A Front	60-contact routing-board mezzanine plus four ground-blade lands and two locating holes. Proposed ASIC contact allocation is separate from CAD. Order: QSH-030-01-L-D-A	All mezzanine contacts: Appendix C.
L1 1uH 5.4A Isat30% Back	Buck energy-storage inductor between SW_CORE and VCORE_REG_1V025. Keep with its converter and local capacitors. Order: XFL4020-102MEC	1: SW_CORE 2: VCORE_REG_1V025
L2 1uH 5.4A Isat30% Front	Buck energy-storage inductor between SW_AUX and VCCAUX_1V8. Keep with its converter and local capacitors. Order: XFL4020-102MEC	1: SW_AUX 2: VCCAUX_1V8
L3 1uH 5.4A Isat30% Front	Buck energy-storage inductor between SW_CFG and VCC_CFG_3V3. Keep with its converter and local capacitors. Order: XFL4020-102MEC	1: SW_CFG 2: VCC_CFG_3V3

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
L4 1uH 5.4A Isat30% Back	Buck energy-storage inductor between SW_ASIC and VCC_ASIC_1V5. Keep with its converter and local capacitors. Order: XFL4020-102MEC	1: SW_ASIC 2: VCC_ASIC_1V5
R1 32.4k 0.1% Back	Upper feedback-divider resistor from the regulated output to FB_CORE. Sets the output together with the lower resistor. Order: TBD - manufacturer/order code not selected	1: VCORE_REG_1V025 2: FB_CORE
R2 69.8k 0.1% Back	Lower feedback-divider resistor from FB_CORE to ground. Retain its ratio with the upper resistor. Order: TBD - manufacturer/order code not selected	1: FB_CORE 2: GND
R3 110k 0.1% Front	Upper feedback-divider resistor from the regulated output to FB_AUX. Sets the output together with the lower resistor. Order: TBD - manufacturer/order code not selected	1: VCCAUX_1V8 2: FB_AUX
R4 69.8k 0.1% Front	Lower feedback-divider resistor from FB_AUX to ground. Retain its ratio with the upper resistor. Order: TBD - manufacturer/order code not selected	1: FB_AUX 2: GND
R5 261k 0.1% Front	Upper feedback-divider resistor from the regulated output to FB_CFG. Sets the output together with the lower resistor. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FB_CFG
R6 69.8k 0.1% Front	Lower feedback-divider resistor from FB_CFG to ground. Retain its ratio with the upper resistor. Order: TBD - manufacturer/order code not selected	1: FB_CFG 2: GND
R7 80.6k 0.1% Back	Upper feedback-divider resistor from the regulated output to FB_ASIC. Sets the output together with the lower resistor. Order: TBD - manufacturer/order code not selected	1: VCC_ASIC_1V5 2: FB_ASIC
R8 69.8k 0.1% Back	Lower feedback-divider resistor from FB_ASIC to ground. Retain its ratio with the upper resistor. Order: TBD - manufacturer/order code not selected	1: FB_ASIC 2: GND
R9 15m 1% 1W Back	15 milliohm series element between the sensed converter output and distributed FPGA core rail. Bulk isolation causes load-dependent voltage drop. Order: WSLP1206R0150FEA	1: VCORE_REG_1V025 2: VCCINT_1V0
R10 100k Back	Pull-up for PG_CORE to LINK_12V. Allows an open-drain power-good output to rise. Order: TBD - manufacturer/order code not selected	1: LINK_12V 2: PG_CORE
R11 100k Front	Pull-up for PG_AUX to LINK_12V. Allows an open-drain power-good output to rise. Order: TBD - manufacturer/order code not selected	1: LINK_12V 2: PG_AUX
R12 100k Front	Pull-up for PGOOD_IO to VCC_CFG_3V3. Allows an open-drain power-good output to rise. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: PGOOD_IO
R100 2.4k Back	Pull-up keeping FLASH_CS_B high when it is not driven. Supports inactive flash-control state. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FLASH_CS_B
R101 4.7k Back	Pull-up keeping FLASH_DQ2 high when it is not driven. Supports inactive flash-control state. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FLASH_DQ2
R102 4.7k Back	Pull-up keeping FLASH_DQ3 high when it is not driven. Supports inactive flash-control state. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FLASH_DQ3
R103 33 Back	Series damping candidate between the FPGA and flash nets. Value and timing still require validation. Order: TBD - manufacturer/order code not selected	1: FPGA_CCLK 2: FLASH_SCLK
R104 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD - manufacturer/order code not selected	1: FPGA_CFG_DQ0 2: FLASH_DQ0

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
R105 33 Back	Series damping candidate between the FPGA and flash nets. Value and timing still require validation. Order: TBD - manufacturer/order code not selected	1: FPGA_CFG_DQ1 2: FLASH_DQ1
R106 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD - manufacturer/order code not selected	1: FPGA_CFG_DQ2 2: FLASH_DQ2
R107 0 Back	Zero-ohm series link / tuning position between the FPGA and flash nets. Value and timing still require validation. Order: TBD - manufacturer/order code not selected	1: FPGA_CFG_DQ3 2: FLASH_DQ3
R108 4.7k Front	Pull-up for the FPGA FPGA_PROGRAM_B control/status node. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FPGA_PROGRAM_B
R109 4.7k Front	Pull-up for the FPGA FPGA_INIT_B control/status node. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FPGA_INIT_B
R110 4.7k Front	Pull-up for the FPGA FPGA_DONE control/status node. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: FPGA_DONE
R111 1k Front	Configuration-mode strap: pull high. The three straps encode M[2:0]=001 in this revision. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: CFG_M0
R112 1k Front	Configuration-mode strap: pull low. The three straps encode M[2:0]=001 in this revision. Order: TBD - manufacturer/order code not selected	1: CFG_M1 2: GND
R113 1k Front	Configuration-mode strap: pull low. The three straps encode M[2:0]=001 in this revision. Order: TBD - manufacturer/order code not selected	1: CFG_M2 2: GND
R114 1k Front	Pulls PUDC_B high to disable configuration-time user-I/O pull-ups. External ASIC safe-state handling remains separate. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: CFG_PUDC_B
R115 10k Front	JTAG input pull-up on JTAG_TMS. Cable drive, voltage and ringing still require qualification. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: JTAG_TMS
R116 10k Front	JTAG input pull-up on JTAG_TDI. Cable drive, voltage and ringing still require qualification. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: JTAG_TDI
R117 10k Front	JTAG input pull-up on JTAG_TCK. Cable drive, voltage and ringing still require qualification. Order: TBD - manufacturer/order code not selected	1: VCC_CFG_3V3 2: JTAG_TCK
R118 33 Back	Series damping candidate on FPGA TDO before J4.18. It is not a qualified long-cable termination. Order: TBD - manufacturer/order code not selected	1: FPGA_TDO 2: JTAG_TDO
R119 33 Front	Series damping candidate at oscillator output; connects OSC_32M_RAW to CLK_32MHZ. Order: TBD - manufacturer/order code not selected	1: OSC_32M_RAW 2: CLK_32MHZ
U1 XC7A100T-CSG324 Front	FPGA logic, memory and I/O. Intended to capture the ASIC outputs and control the downstream link; application assignment and firmware remain open. Order: XC7A100T – speed and temperature grade TBD	All 324 balls: Appendix B.
U2 TPS62135RGXR Back	Core buck converter; senses the local VCORE_REG_1V025 node before R9. Order: TPS62135RGXR	Every IC / oscillator pin: Appendix D.
U3 TPS62135RGXR Front	Auxiliary 1.8 V buck converter; enabled by PG_CORE. Order: TPS62135RGXR	Every IC / oscillator pin: Appendix D.
U4 TPS62135RGXR Front	Configuration / bank 0 / 3.3 V buck converter; enabled by PG_AUX. Order: TPS62135RGXR	Every IC / oscillator pin: Appendix D.

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Ref. / value / face	Why it is here / saved order code	Actual pin-to-net assignment
U5 TPS62135RGXR Back	FPGA ASIC-bank 1.5 V buck converter; enabled by PG_AUX. This is not proof of power delivery to the external ASIC boards. Order: TPS62135RGXR	Every IC / oscillator pin: Appendix D.
U6 MX25L12833FM2I-10G Back	Nonvolatile configuration flash for cold boot. Native candidate retains the 3.3 V part and bank configuration. Order: MX25L12833FM2I-10G	Every IC / oscillator pin: Appendix D.
Y1 ASE-32.000MHZ-L-C-T Front	32 MHz user-reference oscillator. Its output goes through R119 to U1.P17; it is separate from the FPGA configuration clock. Order: ASE-32.000MHZ-L-C-T	Every IC / oscillator pin: Appendix D.

B All 324 FPGA balls

The manufacturer function and bank come from the AMD XC7A100T CSG324 package table. The net/state comes from this exact native PCB. No proposed ASIC FPGA ball has been silently filled in. Assigned rows remain unrouted. Bank “NA” denotes a package field without an I/O-bank number, not an available signal.

Ball	Bank	AMD function	Actual assigned net / state
A1	35	IO_L9N_T1_DQS_AD7N_35	unassigned
A2	NA	GND	GND
A3	35	IO_L8N_T1_AD14N_35	unassigned
A4	35	IO_L8P_T1_AD14P_35	unassigned
A5	35	IO_L3N_T0_DQS_AD5N_35	unassigned
A6	35	IO_L3P_T0_DQS_AD5P_35	unassigned
A7	35	VCCO_35	VCC_ASIC_1V5
A8	16	IO_L12N_T1_MRCC_16	unassigned
A9	16	IO_L14N_T2_SRCC_16	unassigned
A10	16	IO_L14P_T2_SRCC_16	unassigned
A11	15	IO_L4N_T0_15	unassigned
A12	NA	GND	GND
A13	15	IO_L9P_T1_DQS_AD3P_15	unassigned
A14	15	IO_L9N_T1_DQS_AD3N_15	unassigned
A15	15	IO_L8P_T1_AD10P_15	unassigned
A16	15	IO_L8N_T1_AD10N_15	unassigned
A17	15	VCCO_15	VCC_ASIC_1V5
A18	15	IO_L10N_T1_AD11N_15	unassigned
B1	35	IO_L9P_T1_DQS_AD7P_35	unassigned
B2	35	IO_L10N_T1_AD15N_35	unassigned
B3	35	IO_L10P_T1_AD15P_35	unassigned
B4	35	IO_L7N_T1_AD6N_35	unassigned
B5	NA	GND	GND
B6	35	IO_L2N_T0_AD12N_35	unassigned
B7	35	IO_L2P_T0_AD12P_35	unassigned
B8	16	IO_L12P_T1_MRCC_16	unassigned
B9	16	IO_L11N_T1_SRCC_16	unassigned
B10	16	VCCO_16	VCC_CFG_3V3
B11	15	IO_L4P_T0_15	unassigned
B12	15	IO_L3N_T0_DQS_AD1N_15	unassigned
B13	15	IO_L2P_T0_AD8P_15	unassigned
B14	15	IO_L2N_T0_AD8N_15	unassigned
B15	NA	GND	GND
B16	15	IO_L7P_T1_AD2P_15	unassigned
B17	15	IO_L7N_T1_AD2N_15	unassigned
B18	15	IO_L10P_T1_AD11P_15	unassigned
C1	35	IO_L16N_T2_35	unassigned
C2	35	IO_L16P_T2_35	unassigned
C3	35	VCCO_35	VCC_ASIC_1V5
C4	35	IO_L7P_T1_AD6P_35	unassigned
C5	35	IO_L1N_T0_AD4N_35	unassigned
C6	35	IO_L1P_T0_AD4P_35	unassigned
C7	35	IO_L4N_T0_35	unassigned
C8	NA	GND	GND
C9	16	IO_L11P_T1_SRCC_16	unassigned
C10	16	IO_L13N_T2_MRCC_16	unassigned

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Ball	Bank	AMD function	Actual assigned net / state
C11	16	IO_L13P_T2_MRCC_16	unassigned
C12	15	IO_L3P_T0_DQS_AD1P_15	unassigned
C13	15	VCCO_15	VCC_ASIC_1V5
C14	15	IO_L1N_T0_AD0N_15	unassigned
C15	15	IO_L12N_T1_MRCC_15	unassigned
C16	15	IO_L20P_T3_A20_15	unassigned
C17	15	IO_L20N_T3_A19_15	unassigned
C18	NA	GND	GND
D1	NA	GND	GND
D2	35	IO_L14N_T2_SRCC_35	unassigned
D3	35	IO_L12N_T1_MRCC_35	unassigned
D4	35	IO_L11N_T1_SRCC_35	unassigned
D5	35	IO_L11P_T1_SRCC_35	unassigned
D6	35	VCCO_35	VCC_ASIC_1V5
D7	35	IO_L6N_T0_VREF_35	unassigned
D8	35	IO_L4P_T0_35	unassigned
D9	16	IO_L6N_T0_VREF_16	unassigned
D10	16	IO_L19N_T3_VREF_16	unassigned
D11	NA	GND	GND
D12	15	IO_L6P_T0_15	unassigned
D13	15	IO_L6N_T0_VREF_15	unassigned
D14	15	IO_L1P_T0_AD0P_15	unassigned
D15	15	IO_L12P_T1_MRCC_15	unassigned
D16	15	VCCO_15	VCC_ASIC_1V5
D17	15	IO_L16N_T2_A27_15	unassigned
D18	15	IO_L21N_T3_DQS_A18_15	unassigned
E1	35	IO_L18N_T2_35	unassigned
E2	35	IO_L14P_T2_SRCC_35	unassigned
E3	35	IO_L12P_T1_MRCC_35	unassigned
E4	NA	GND	GND
E5	35	IO_L5N_T0_AD13N_35	unassigned
E6	35	IO_L5P_T0_AD13P_35	unassigned
E7	35	IO_L6P_T0_35	unassigned
E8	0	VCCBATT_0	GND
E9	0	CCLK_0	FPGA_CCLK
E10	0	TCK_0	JTAG_TCK
E11	0	TDI_0	JTAG_TDI
E12	0	TMS_0	JTAG_TMS
E13	0	TDO_0	FPGA_TDO
E14	NA	GND	GND
E15	15	IO_L11P_T1_SRCC_15	unassigned
E16	15	IO_L11N_T1_SRCC_15	unassigned
E17	15	IO_L16P_T2_A28_15	unassigned
E18	15	IO_L21P_T3_DQS_15	unassigned
F1	35	IO_L18P_T2_35	unassigned
F2	35	VCCO_35	VCC_ASIC_1V5
F3	35	IO_L13N_T2_MRCC_35	unassigned
F4	35	IO_L13P_T2_MRCC_35	unassigned
F5	35	IO_0_35	unassigned
F6	35	IO_L19N_T3_VREF_35	unassigned
F7	NA	GND	GND
F8	NA	VCCINT	VCCINT_1V0
F9	NA	GND	GND
F10	NA	VCCBRAM	VCCINT_1V0

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Ball	Bank	AMD function	Actual assigned net / state
F11	NA	GND	GND
F12	NA	VCCAUX	VCCAUX_1V8
F13	15	IO_L5P_T0_AD9P_15	unassigned
F14	15	IO_L5N_T0_AD9N_15	unassigned
F15	15	IO_L14P_T2_SRCC_15	unassigned
F16	15	IO_L14N_T2_SRCC_15	unassigned
F17	NA	GND	GND
F18	15	IO_L22N_T3_A16_15	unassigned
G1	35	IO_L17N_T2_35	unassigned
G2	35	IO_L15N_T2_DQS_35	unassigned
G3	35	IO_L20N_T3_35	unassigned
G4	35	IO_L20P_T3_35	unassigned
G5	35	VCCO_35	VCC_ASIC_1V5
G6	35	IO_L19P_T3_35	unassigned
G7	NA	VCCINT	VCCINT_1V0
G8	NA	GND	GND
G9	NA	VCCINT	VCCINT_1V0
G10	NA	GND	GND
G11	NA	VCCBRAM	VCCINT_1V0
G12	NA	GND	GND
G13	15	IO_0_15	unassigned
G14	15	IO_L15N_T2_DQS_ADV_B_15	unassigned
G15	15	VCCO_15	VCC_ASIC_1V5
G16	15	IO_L13N_T2_MRCC_15	unassigned
G17	15	IO_L18N_T2_A23_15	unassigned
G18	15	IO_L22P_T3_A17_15	unassigned
H1	35	IO_L17P_T2_35	unassigned
H2	35	IO_L15P_T2_DQS_35	unassigned
H3	NA	GND	GND
H4	35	IO_L21N_T3_DQS_35	unassigned
H5	35	IO_L24N_T3_35	unassigned
H6	35	IO_L24P_T3_35	unassigned
H7	NA	GND	GND
H8	NA	VCCINT	VCCINT_1V0
H9	0	GNDADC_0	GND
H10	0	VCCADC_0	VCCAUX_1V8
H11	NA	GND	GND
H12	NA	VCCAUX	VCCAUX_1V8
H13	NA	GND	GND
H14	15	IO_L15P_T2_DQS_15	unassigned
H15	15	IO_L19N_T3_A21_VREF_15	unassigned
H16	15	IO_L13P_T2_MRCC_15	unassigned
H17	15	IO_L18P_T2_A24_15	unassigned
H18	15	VCCO_15	VCC_ASIC_1V5
J1	35	VCCO_35	VCC_ASIC_1V5
J2	35	IO_L22N_T3_35	unassigned
J3	35	IO_L22P_T3_35	unassigned
J4	35	IO_L21P_T3_DQS_35	unassigned
J5	35	IO_25_35	unassigned
J6	NA	GND	GND
J7	NA	VCCINT	VCCINT_1V0
J8	NA	GND	GND
J9	0	VREFN_0	GND
J10	0	VP_0	GND

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Ball	Bank	AMD function	Actual assigned net / state
J11	NA	VCCINT	VCCINT_1V0
J12	NA	GND	GND
J13	15	IO_L17N_T2_A25_15	unassigned
J14	15	IO_L19P_T3_A22_15	unassigned
J15	15	IO_L24N_T3_RS0_15	unassigned
J16	NA	GND	GND
J17	15	IO_L23P_T3_FOE_B_15	unassigned
J18	15	IO_L23N_T3_FWE_B_15	unassigned
K1	35	IO_L23N_T3_35	unassigned
K2	35	IO_L23P_T3_35	unassigned
K3	34	IO_L2P_T0_34	unassigned
K4	34	VCCO_34	VCC_ASIC_1V5
K5	34	IO_L5P_T0_34	unassigned
K6	34	IO_0_34	unassigned
K7	NA	GND	GND
K8	NA	VCCINT	VCCINT_1V0
K9	0	VN_0	GND
K10	0	VREFP_0	GND
K11	NA	GND	GND
K12	NA	VCCAUX	VCCAUX_1V8
K13	15	IO_L17P_T2_A26_15	unassigned
K14	15	VCCO_15	VCC_ASIC_1V5
K15	15	IO_L24P_T3_RS1_15	unassigned
K16	15	IO_25_15	unassigned
K17	14	IO_L1P_T0_D00_MOSI_14	FPGA_CFG_DQ0
K18	14	IO_L1N_T0_D01_DIN_14	FPGA_CFG_DQ1
L1	34	IO_L1P_T0_34	unassigned
L2	NA	GND	GND
L3	34	IO_L2N_T0_34	unassigned
L4	34	IO_L5N_T0_34	unassigned
L5	34	IO_L6N_T0_VREF_34	unassigned
L6	34	IO_L6P_T0_34	unassigned
L7	NA	VCCINT	VCCINT_1V0
L8	NA	GND	GND
L9	0	DXN_0	intentional NC
L10	0	DXP_0	intentional NC
L11	NA	VCCINT	VCCINT_1V0
L12	NA	GND	GND
L13	14	IO_L6P_T0_FCS_B_14	FLASH_CS_B
L14	14	IO_L2P_T0_D02_14	FPGA_CFG_DQ2
L15	14	IO_L3P_T0_DQS_PUDC_B_14	CFG_PUDC_B
L16	14	IO_L3N_T0_DQS_EMCCLK_14	unassigned
L17	14	VCCO_14	VCC_CFG_3V3
L18	14	IO_L4P_T0_D04_14	unassigned
M1	34	IO_L1N_T0_34	unassigned
M2	34	IO_L4N_T0_34	unassigned
M3	34	IO_L4P_T0_34	unassigned
M4	34	IO_L16P_T2_34	unassigned
M5	NA	GND	GND
M6	34	IO_L18P_T2_34	unassigned
M7	NA	GND	GND
M8	NA	VCCINT	VCCINT_1V0
M9	NA	GND	GND
M10	NA	VCCINT	VCCINT_1V0

Continued on the next page

Ball	Bank	AMD function	Actual assigned net / state
M11	NA	GND	GND
M12	NA	VCCAUX	VCCAUX_1V8
M13	14	IO_L6N_T0_D08_VREF_14	unassigned
M14	14	IO_L2N_T0_D03_14	FPGA_CFG_DQ3
M15	NA	GND	GND
M16	14	IO_L10P_T1_D14_14	unassigned
M17	14	IO_L10N_T1_D15_14	unassigned
M18	14	IO_L4N_T0_D05_14	unassigned
N1	34	IO_L3N_T0_DQS_34	unassigned
N2	34	IO_L3P_T0_DQS_34	unassigned
N3	34	VCCO_34	VCC_ASIC_1V5
N4	34	IO_L16N_T2_34	unassigned
N5	34	IO_L13P_T2_MRCC_34	unassigned
N6	34	IO_L18N_T2_34	unassigned
N7	NA	VCCINT	VCCINT_1V0
N8	NA	GND	GND
N9	NA	VCCINT	VCCINT_1V0
N10	NA	GND	GND
N11	NA	VCCINT	VCCINT_1V0
N12	NA	GND	GND
N13	14	VCCO_14	VCC_CFG_3V3
N14	14	IO_L8P_T1_D11_14	unassigned
N15	14	IO_L11P_T1_SRCC_14	unassigned
N16	14	IO_L11N_T1_SRCC_14	unassigned
N17	14	IO_L9P_T1_DQS_14	unassigned
N18	NA	GND	GND
P1	NA	GND	GND
P2	34	IO_L15P_T2_DQS_34	unassigned
P3	34	IO_L14N_T2_SRCC_34	unassigned
P4	34	IO_L14P_T2_SRCC_34	unassigned
P5	34	IO_L13N_T2_MRCC_34	unassigned
P6	34	VCCO_34	VCC_ASIC_1V5
P7	0	INIT_B_0	FPGA_INIT_B
P8	0	CFGBVS_0	VCC_CFG_3V3
P9	0	PROGRAM_B_0	FPGA_PROGRAM_B
P10	0	DONE_0	FPGA_DONE
P11	0	M2_0	CFG_M2
P12	0	M0_0	CFG_M0
P13	0	M1_0	CFG_M1
P14	14	IO_L8N_T1_D12_14	unassigned
P15	14	IO_L13P_T2_MRCC_14	unassigned
P16	14	VCCO_14	VCC_CFG_3V3
P17	14	IO_L12P_T1_MRCC_14	CLK_32MHZ
P18	14	IO_L9N_T1_DQS_D13_14	unassigned
R1	34	IO_L17P_T2_34	unassigned
R2	34	IO_L15N_T2_DQS_34	unassigned
R3	34	IO_L11P_T1_SRCC_34	unassigned
R4	NA	GND	GND
R5	34	IO_L19N_T3_VREF_34	unassigned
R6	34	IO_L19P_T3_34	unassigned
R7	34	IO_L23P_T3_34	unassigned
R8	34	IO_L24P_T3_34	unassigned
R9	0	VCCO_0	VCC_CFG_3V3
R10	14	IO_25_14	unassigned

Continued on the next page

Ball	Bank	AMD function	Actual assigned net / state
R11	14	IO_0_14	unassigned
R12	14	IO_L5P_T0_D06_14	unassigned
R13	14	IO_L5N_T0_D07_14	unassigned
R14	NA	GND	GND
R15	14	IO_L13N_T2_MRCC_14	unassigned
R16	14	IO_L15P_T2_DQS_RDWR_B_14	unassigned
R17	14	IO_L12N_T1_MRCC_14	unassigned
R18	14	IO_L7P_T1_D09_14	unassigned
T1	34	IO_L17N_T2_34	unassigned
T2	34	VCCO_34	VCC_ASIC_1V5
T3	34	IO_L11N_T1_SRCC_34	unassigned
T4	34	IO_L12N_T1_MRCC_34	unassigned
T5	34	IO_L12P_T1_MRCC_34	unassigned
T6	34	IO_L23N_T3_34	unassigned
T7	NA	GND	GND
T8	34	IO_L24N_T3_34	unassigned
T9	14	IO_L24P_T3_A01_D17_14	unassigned
T10	14	IO_L24N_T3_A00_D16_14	unassigned
T11	14	IO_L19P_T3_A10_D26_14	unassigned
T12	14	VCCO_14	VCC_CFG_3V3
T13	14	IO_L23P_T3_A03_D19_14	unassigned
T14	14	IO_L14P_T2_SRCC_14	unassigned
T15	14	IO_L14N_T2_SRCC_14	unassigned
T16	14	IO_L15N_T2_DQS_DOUT_CSO_B_14	unassigned
T17	NA	GND	GND
T18	14	IO_L7N_T1_D10_14	unassigned
U1	34	IO_L7P_T1_34	unassigned
U2	34	IO_L9P_T1_DQS_34	unassigned
U3	34	IO_L8N_T1_34	unassigned
U4	34	IO_L8P_T1_34	unassigned
U5	34	VCCO_34	VCC_ASIC_1V5
U6	34	IO_L22N_T3_34	unassigned
U7	34	IO_L22P_T3_34	unassigned
U8	34	IO_25_34	unassigned
U9	34	IO_L21P_T3_DQS_34	unassigned
U10	NA	GND	GND
U11	14	IO_L19N_T3_A09_D25_VREF_14	unassigned
U12	14	IO_L20P_T3_A08_D24_14	unassigned
U13	14	IO_L23N_T3_A02_D18_14	unassigned
U14	14	IO_L22P_T3_A05_D21_14	unassigned
U15	14	VCCO_14	VCC_CFG_3V3
U16	14	IO_L18P_T2_A12_D28_14	unassigned
U17	14	IO_L17P_T2_A14_D30_14	unassigned
U18	14	IO_L17N_T2_A13_D29_14	unassigned
V1	34	IO_L7N_T1_34	unassigned
V2	34	IO_L9N_T1_DQS_34	unassigned
V3	NA	GND	GND
V4	34	IO_L10N_T1_34	unassigned
V5	34	IO_L10P_T1_34	unassigned
V6	34	IO_L20N_T3_34	unassigned
V7	34	IO_L20P_T3_34	unassigned
V8	34	VCCO_34	VCC_ASIC_1V5
V9	34	IO_L21N_T3_DQS_34	unassigned
V10	14	IO_L21P_T3_DQS_14	unassigned

Continued on the next page

Ball	Bank	AMD function	Actual assigned net / state
V11	14	IO_L21N_T3_DQS_A06_D22_14	unassigned
V12	14	IO_L20N_T3_A07_D23_14	unassigned
V13	NA	GND	GND
V14	14	IO_L22N_T3_A04_D20_14	unassigned
V15	14	IO_L16P_T2_CSI_B_14	unassigned
V16	14	IO_L16N_T2_A15_D31_14	unassigned
V17	14	IO_L18N_T2_A11_D27_14	unassigned
V18	14	VCCO_14	VCC_CFG_3V3

C External connector contacts

C.1 J4: custom power, JTAG and reserved link

This uses a micro-HDMI-style receptacle mechanically, but the proposed pin map is custom. J4.19 is assigned 12 V; it must not be connected to ordinary HDMI equipment. No cable, connector current rating, receiver/carrier or input protection has been qualified for this assignment. The eight reserved data contacts have no functional PCB nets.

Pin	Function	Actual assigned net / state	Physical pads
1	VTREF	VCC_CFG_3V3	1
2	JTAG_TMS	JTAG_TMS	1
3	PAIR0_P_RESERVED	unassigned	1
4	GND	GND	1
5	PAIR0_N_RESERVED	unassigned	1
6	PAIR1_P_RESERVED	unassigned	1
7	GND	GND	1
8	PAIR1_N_RESERVED	unassigned	1
9	PAIR2_P_RESERVED	unassigned	1
10	GND	GND	1
11	PAIR2_N_RESERVED	unassigned	1
12	PAIR3_P_RESERVED	unassigned	1
13	GND	GND	1
14	PAIR3_N_RESERVED	unassigned	1
15	JTAG_TDI	JTAG_TDI	1
16	GND	GND	1
17	JTAG_TCK	JTAG_TCK	1
18	JTAG_TDO	JTAG_TDO	1
19	12V	LINK_12V	1
SH	GND	GND	4

C.2 J5 and J6: actual state versus 117-signal proposal

Each QSH-030 connector has 60 numbered signal contacts. All of these are **unassigned in the PCB**. The separate proposal below is included so the intended destination is visible; it is not a completed electrical pinout. No proposal has an implemented FPGA-ball connection. The signal-direction column is the proposal's direction relative to the FPGA, not a verified ASIC electrical classification.

Both connectors also have four physical ground-blade lands sharing pad number G. Those lands are assigned to GND, but there is no ground copper in this revision. Two unnumbered non-plated locating holes per connector provide no electrical connection.

Contact	Actual PCB state	Separate proposed signal	Proposed direction
J5.1	unassigned	ASIC1_DATA1	Input
J5.2	unassigned	ASIC1_DATA2	Input
J5.3	unassigned	ASIC1_DATA3	Input
J5.4	unassigned	ASIC1_DATA4	Input
J5.5	unassigned	ASIC1_DATA5	Input
J5.6	unassigned	ASIC1_DATA6	Input
J5.7	unassigned	ASIC1_DATA7	Input
J5.8	unassigned	ASIC1_DATA8	Input
J5.9	unassigned	ASIC1_CLK32MHz_Out	Input

Continued on the next page

Contact	Actual PCB state	Separate proposed signal	Proposed direction
J5.10	unassigned	ASIC1_READ	Input
J5.11	unassigned	ASIC1_SYNC	Input
J5.12	unassigned	ASIC2_DATA1	Input
J5.13	unassigned	ASIC2_DATA2	Input
J5.14	unassigned	ASIC2_DATA3	Input
J5.15	unassigned	ASIC2_DATA4	Input
J5.16	unassigned	ASIC2_DATA5	Input
J5.17	unassigned	ASIC2_DATA6	Input
J5.18	unassigned	ASIC2_DATA7	Input
J5.19	unassigned	ASIC2_DATA8	Input
J5.20	unassigned	ASIC2_CLK32MHz_Out	Input
J5.21	unassigned	ASIC2_READ	Input
J5.22	unassigned	ASIC2_SYNC	Input
J5.23	unassigned	ASIC3_DATA1	Input
J5.24	unassigned	ASIC3_DATA2	Input
J5.25	unassigned	ASIC3_DATA3	Input
J5.26	unassigned	ASIC3_DATA4	Input
J5.27	unassigned	ASIC3_DATA5	Input
J5.28	unassigned	ASIC3_DATA6	Input
J5.29	unassigned	ASIC3_DATA7	Input
J5.30	unassigned	ASIC3_DATA8	Input
J5.31	unassigned	ASIC3_CLK32MHz_Out	Input
J5.32	unassigned	ASIC3_READ	Input
J5.33	unassigned	ASIC3_SYNC	Input
J5.34	unassigned	ASIC4_DATA1	Input
J5.35	unassigned	ASIC4_DATA2	Input
J5.36	unassigned	ASIC4_DATA3	Input
J5.37	unassigned	ASIC4_DATA4	Input
J5.38	unassigned	ASIC4_DATA5	Input
J5.39	unassigned	ASIC4_DATA6	Input
J5.40	unassigned	ASIC4_DATA7	Input
J5.41	unassigned	ASIC4_DATA8	Input
J5.42	unassigned	ASIC4_CLK32MHz_Out	Input
J5.43	unassigned	ASIC4_READ	Input
J5.44	unassigned	ASIC4_SYNC	Input
J5.45	unassigned	CHIP_RESET_SHARED	Output
J5.46	unassigned	AC_IN_SHARED	TBD: verify circuit role
J5.47	unassigned	IMP_TST_SHARED	TBD: verify circuit role
J5.48	unassigned	FE_RESET_SHARED	Output
J5.49	unassigned	SPI_LATCH_SHARED	Output
J5.50	unassigned	STIM_CLK_SHARED	Output
J5.51	unassigned	STIM_START_SHARED	Output
J5.52	unassigned	STIM_EN_SHARED	Output
J5.53	unassigned	STIM_CHB_SHARED	Output
J5.54	unassigned	BOARD1_CLK	Output
J5.55	unassigned	BOARD2_CLK	Output
J5.56	unassigned	BOARD3_CLK	Output
J5.57	unassigned	BOARD4_CLK	Output
J5.58	unassigned	BOARD1_SPI_CLK	Output
J5.59	unassigned	BOARD2_SPI_CLK	Output
J5.60	unassigned	BOARD3_SPI_CLK	Output
J5.G	GND	GND	Ground reference
J6.1	unassigned	ASIC5_DATA1	Input
J6.2	unassigned	ASIC5_DATA2	Input

Continued on the next page

Contact	Actual PCB state	Separate proposed signal	Proposed direction
J6.3	unassigned	ASIC5_DATA3	Input
J6.4	unassigned	ASIC5_DATA4	Input
J6.5	unassigned	ASIC5_DATA5	Input
J6.6	unassigned	ASIC5_DATA6	Input
J6.7	unassigned	ASIC5_DATA7	Input
J6.8	unassigned	ASIC5_DATA8	Input
J6.9	unassigned	ASIC5_CLK32MHz_Out	Input
J6.10	unassigned	ASIC5_READ	Input
J6.11	unassigned	ASIC5_SYNC	Input
J6.12	unassigned	ASIC6_DATA1	Input
J6.13	unassigned	ASIC6_DATA2	Input
J6.14	unassigned	ASIC6_DATA3	Input
J6.15	unassigned	ASIC6_DATA4	Input
J6.16	unassigned	ASIC6_DATA5	Input
J6.17	unassigned	ASIC6_DATA6	Input
J6.18	unassigned	ASIC6_DATA7	Input
J6.19	unassigned	ASIC6_DATA8	Input
J6.20	unassigned	ASIC6_CLK32MHz_Out	Input
J6.21	unassigned	ASIC6_READ	Input
J6.22	unassigned	ASIC6_SYNC	Input
J6.23	unassigned	ASIC7_DATA1	Input
J6.24	unassigned	ASIC7_DATA2	Input
J6.25	unassigned	ASIC7_DATA3	Input
J6.26	unassigned	ASIC7_DATA4	Input
J6.27	unassigned	ASIC7_DATA5	Input
J6.28	unassigned	ASIC7_DATA6	Input
J6.29	unassigned	ASIC7_DATA7	Input
J6.30	unassigned	ASIC7_DATA8	Input
J6.31	unassigned	ASIC7_CLK32MHz_Out	Input
J6.32	unassigned	ASIC7_READ	Input
J6.33	unassigned	ASIC7_SYNC	Input
J6.34	unassigned	ASIC8_DATA1	Input
J6.35	unassigned	ASIC8_DATA2	Input
J6.36	unassigned	ASIC8_DATA3	Input
J6.37	unassigned	ASIC8_DATA4	Input
J6.38	unassigned	ASIC8_DATA5	Input
J6.39	unassigned	ASIC8_DATA6	Input
J6.40	unassigned	ASIC8_DATA7	Input
J6.41	unassigned	ASIC8_DATA8	Input
J6.42	unassigned	ASIC8_CLK32MHz_Out	Input
J6.43	unassigned	ASIC8_READ	Input
J6.44	unassigned	ASIC8_SYNC	Input
J6.45	unassigned	BOARD4_SPI_CLK	Output
J6.46	unassigned	STIM_CHIP1_SPI_DL	Output
J6.47	unassigned	STIM_CHIP1_SPI_DR	Output
J6.48	unassigned	STIM_CHIP2_SPI_DL	Output
J6.49	unassigned	STIM_CHIP2_SPI_DR	Output
J6.50	unassigned	STIM_CHIP3_SPI_DL	Output
J6.51	unassigned	STIM_CHIP3_SPI_DR	Output
J6.52	unassigned	STIM_CHIP4_SPI_DL	Output
J6.53	unassigned	STIM_CHIP4_SPI_DR	Output
J6.54	unassigned	NONSTIM_BOARD1_SPI_DL	Output
J6.55	unassigned	NONSTIM_BOARD1_SPI_DR	Output
J6.56	unassigned	NONSTIM_BOARD2_SPI_DL	Output

Continued on the next page

Contact	Actual PCB state	Separate proposed signal	Proposed direction
J6.57	unassigned	NONSTIM_BOARD2_SPI_DR	Output
J6.58	unassigned	RESERVED_UNCONNECTED	Reserved
J6.59	unassigned	RESERVED_UNCONNECTED	Reserved
J6.60	unassigned	RESERVED_UNCONNECTED	Reserved
J6.G	GND	GND	Ground reference

D All converter, flash and oscillator pins

These rows complete the non-FPGA IC and oscillator pin coverage. FB2 pins are intentional NC; every other assigned row is still unrouted. Grounded VSEL and powered EN remain distinct pin functions. Flash and oscillator functions follow the selected part interpretation; exact assembly order codes must still be qualified with their footprints.

Endpoint	Pin function	Actual assigned net / state
U2.1	VIN	LINK_12V
U2.2	SW	SW_CORE
U2.3	GND	GND
U2.4	FB2	intentional NC
U2.5	FB	FB_CORE
U2.6	VOS	VCORE_REG_1V025
U2.7	PG	PG_CORE
U2.8	EN	LINK_12V
U2.9	SS/TR	SS_CORE
U2.10	MODE	LINK_12V
U2.11	VSEL	GND
U3.1	VIN	LINK_12V
U3.2	SW	SW_AUX
U3.3	GND	GND
U3.4	FB2	intentional NC
U3.5	FB	FB_AUX
U3.6	VOS	VCCAUX_1V8
U3.7	PG	PG_AUX
U3.8	EN	PG_CORE
U3.9	SS/TR	SS_AUX
U3.10	MODE	LINK_12V
U3.11	VSEL	GND
U4.1	VIN	LINK_12V
U4.2	SW	SW_CFG
U4.3	GND	GND
U4.4	FB2	intentional NC
U4.5	FB	FB_CFG
U4.6	VOS	VCC_CFG_3V3
U4.7	PG	PGOOD_IO
U4.8	EN	PG_AUX
U4.9	SS/TR	SS_CFG
U4.10	MODE	LINK_12V
U4.11	VSEL	GND
U5.1	VIN	LINK_12V
U5.2	SW	SW_ASIC
U5.3	GND	GND
U5.4	FB2	intentional NC
U5.5	FB	FB_ASIC
U5.6	VOS	VCC_ASIC_1V5
U5.7	PG	PGOOD_IO
U5.8	EN	PG_AUX
U5.9	SS/TR	SS_ASIC
U5.10	MODE	LINK_12V
U5.11	VSEL	GND
U6.1	CS (active low)	FLASH_CS_B
U6.2	SO/SIO1	FLASH_DQ1

Continued on the next page

Endpoint	Pin function	Actual assigned net / state
U6.3	WP (active low)/SIO2	FLASH_DQ2
U6.4	GND	GND
U6.5	SI/SIO0	FLASH_DQ0
U6.6	SCLK	FLASH_SCLK
U6.7	RESET (active low)/SIO3	FLASH_DQ3
U6.8	VCC	VCC_CFG_3V3
Y1.1	EN	VCC_CFG_3V3
Y1.2	GND	GND
Y1.3	OUT	OSC_32M_RAW
Y1.4	Vdd	VCC_CFG_3V3

E Every assigned net and its complete peer list

Endpoints in the same row have the same assigned KiCad net name. They are *not connected by copper* in this placement. Series components intentionally create different nets on their two ends: for example, R119 joins OSC_32M_RAW and CLK_32MHZ once its pads are routed and it is populated. “x4” means four physical lands share that endpoint name.

Assigned net	Every endpoint on this net
CFG_M0	R111.2, U1.P12
CFG_M1	R112.1, U1.P13
CFG_M2	R113.1, U1.P11
CFG_PUDC_B	R114.2, U1.L15
CLK_32MHZ	R119.2, U1.P17
FB_ASIC	R7.2, R8.1, U5.5
FB_AUX	R3.2, R4.1, U3.5
FB_CFG	R5.2, R6.1, U4.5
FB_CORE	R1.2, R2.1, U2.5
FLASH_CS_B	R100.2, U1.L13, U6.1
FLASH_DQ0	R104.2, U6.5
FLASH_DQ1	R105.2, U6.2
FLASH_DQ2	R101.2, R106.2, U6.3
FLASH_DQ3	R102.2, R107.2, U6.7
FLASH_SCLK	R103.2, U6.6
FPGA_CCLK	R103.1, U1.E9
FPGA_CFG_DQ0	R104.1, U1.K17
FPGA_CFG_DQ1	R105.1, U1.K18
FPGA_CFG_DQ2	R106.1, U1.L14
FPGA_CFG_DQ3	R107.1, U1.M14
FPGA_DONE	R110.2, U1.P10
FPGA_INIT_B	R109.2, U1.P7
FPGA_PROGRAM_B	R108.2, U1.P9
FPGA_TDO	R118.1, U1.E13
GND	C1.2, C2.2, C3.2, C4.2, C5.2, C6.2, C7.2, C8.2, C9.2, C10.2, C11.2, C12.2, C13.2, C14.2, C15.2, C16.2, C17.2, C18.2
GND (continued)	C20.2, C21.2, C22.2, C23.2, C24.2, C25.2, C26.2, C27.2, C28.2, C29.2, C30.2, C31.2, C32.2, C33.2, C34.2, C35.2, C36.2, C37.2
GND (continued)	C40.2, C41.2, C42.2, C43.2, C44.2, C45.2, C46.2, C50.2, C51.2, C52.2, C53.2, C54.2, C55.2, C56.2, C57.2, C58.2, C59.2, C60.2
GND (continued)	C61.2, C62.2, C63.2, C64.2, C65.2, C66.2, C67.2, C68.2, C69.2, C70.2, C71.2, C72.2, C73.2, C74.2, C75.2, C76.2, C77.2, C78.2
GND (continued)	C79.2, C80.2, C81.2, C82.2, C90.2, C91.2, C100.2, C101.2, C102.2, C103.2, J4.4, J4.7, J4.10, J4.13, J4.16, J4.SH(x4), J5.G(x4), J6.G(x4)
GND (continued)	R2.2, R4.2, R6.2, R8.2, R112.2, R113.2, U1.A2, U1.A12, U1.B5, U1.B15, U1.C8, U1.C18, U1.D1, U1.D11, U1.E4, U1.E8, U1.E14, U1.F7
GND (continued)	U1.F9, U1.F11, U1.F17, U1.G8, U1.G10, U1.G12, U1.H3, U1.H7, U1.H9, U1.H11, U1.H13, U1.J6, U1.J8, U1.J9, U1.J10, U1.J12, U1.J16, U1.K7
GND (continued)	U1.K9, U1.K10, U1.K11, U1.L2, U1.L8, U1.L12, U1.M5, U1.M7, U1.M9, U1.M11, U1.M15, U1.N8, U1.N10, U1.N12, U1.N18, U1.P1, U1.R4, U1.R14
GND (continued)	U1.T7, U1.T17, U1.U10, U1.V3, U1.V13, U2.3, U2.11, U3.3, U3.11, U4.3, U4.11, U5.3, U5.11, U6.4, Y1.2
JTAG_TCK	J4.17, R117.2, U1.E10
JTAG_TDI	J4.15, R116.2, U1.E11
JTAG_TDO	J4.18, R118.2
JTAG_TMS	J4.2, R115.2, U1.E12

Continued on the next page

Assigned net	Every endpoint on this net
LINK_12V	C1.1, C2.1, C3.1, C4.1, C7.1, C8.1, C11.1, C12.1, C15.1, C16.1, J4.19, R10.1, R11.1, U2.1, U2.8, U2.10, U3.1, U3.10
LINK_12V (continued)	U4.1, U4.10, U5.1, U5.10
OSC_32M_RAW	R119.1, Y1.3
PGOOD_IO	R12.2, U4.7, U5.7
PG_AUX	R11.2, U3.7, U4.8, U5.8
PG_CORE	R10.2, U2.7, U3.8
SS_ASIC	C18.1, U5.9
SS_AUX	C10.1, U3.9
SS_CFG	C14.1, U4.9
SS_CORE	C6.1, U2.9
SW_ASIC	L4.1, U5.2
SW_AUX	L2.1, U3.2
SW_CFG	L3.1, U4.2
SW_CORE	L1.1, U2.2
VCCAUX_1V8	C9.1, C40.1, C41.1, C42.1, C43.1, C44.1, C45.1, C46.1, C90.1, C91.1, L2.2, R3.1, U1.F12, U1.H10, U1.H12, U1.K12, U1.M12, U3.6
VCCINT_1V0	C20.1, C21.1, C22.1, C23.1, C24.1, C25.1, C26.1, C27.1, C28.1, C29.1, C30.1, C31.1, C32.1, C33.1, C34.1, C35.1, C36.1, C37.1
VCCINT_1V0 (continued)	R9.2, U1.F8, U1.F10, U1.G7, U1.G9, U1.G11, U1.H8, U1.J7, U1.J11, U1.K8, U1.L7, U1.L11, U1.M8, U1.M10, U1.N7, U1.N9, U1.N11
VCC_ASIC_1V5	C17.1, C64.1, C65.1, C66.1, C67.1, C68.1, C69.1, C70.1, C71.1, C72.1, C73.1, C74.1, C75.1, C76.1, C77.1, C78.1, C79.1, C80.1
VCC_ASIC_1V5 (continued)	C81.1, C82.1, L4.2, R7.1, U1.A7, U1.A17, U1.C3, U1.C13, U1.D6, U1.D16, U1.F2, U1.G5, U1.G15, U1.H18, U1.J1, U1.K4, U1.K14, U1.N3
VCC_ASIC_1V5 (continued)	U1.P6, U1.T2, U1.U5, U1.V8, U5.6
VCC_CFG_3V3	C13.1, C50.1, C51.1, C52.1, C53.1, C54.1, C55.1, C56.1, C57.1, C58.1, C59.1, C60.1, C61.1, C62.1, C63.1, C100.1, C101.1, C102.1
VCC_CFG_3V3 (continued)	C103.1, J4.1, L3.2, R5.1, R12.1, R100.1, R101.1, R102.1, R108.1, R109.1, R110.1, R111.1, R114.1, R115.1, R116.1, R117.1, U1.B10, U1.L17
VCC_CFG_3V3 (continued)	U1.N13, U1.P8, U1.P16, U1.R9, U1.T12, U1.U15, U1.V18, U4.6, U6.8, Y1.1, Y1.4
VCORE_REG_1V025	C5.1, L1.2, R1.1, R9.1, U2.6

F Evidence, reproducibility and sources

F.1 Native extraction and completeness checks

The native board was reopened in KiCad's Python interface after it was saved. All assigned pins were compared with the inherited core component ledger. The final placement was also checked by the KiCad command-line DRC engine using the copied project rules. There are zero physical DRC violations, 383 unconnected items and no ignored/excluded findings in the delivered report. No integrated schematic exists for this placement, so a schematic/ERC or full schematic-to-PCB parity pass is not claimed.

Appendix A covers 236 endpoints on 118 two-terminal components; B covers 324 FPGA balls; C covers 142 distinct connector endpoints; D covers 56 other IC/oscillator pins. Together these are **758 unique electrical endpoints**, with no duplicate or missing endpoint. The raw physical-pad CSV additionally retains all 807 pad records, including duplicate numbered lands, paste-only apertures and locating holes.

Native board SHA-256:

e2aa37d7b569608729ed44e2d7946d30d108c2124e244e9707d49e6ca25df3de

The source bundle includes the editable LaTeX, the native placement figure, four CSV ledgers, the exact JSON readback, independent DRC output and report-generation instructions. The native KiCad ZIP includes the project and custom libraries. The earlier 37.5 × 36 mm revision is preserved separately.

F.2 Primary references and local design evidence

- [1] Texas Instruments, *TPS62135 / TPS621351 datasheet*, pin table and layout guidance. <https://www.ti.com/lit/ds/symlink/tps62135.pdf>
- [2] AMD, *7 Series FPGAs Configuration User Guide (UG470)*. https://docs.amd.com/v/u/en-US/ug470_7Series_Config
- [3] AMD, *7 Series Packaging and Pinout (UG475)* and the saved xc7a100tcsg324pkg.csv package table. https://docs.amd.com/v/u/en-US/ug475_7Series_Pkg_Pinout
- [4] Macronix, *MX25L12833F datasheet*, PM2517, selected-device pin functions; manufacturer-authored copy. https://www.mouser.com/datasheet/2/819/MX25L12833F_2c_3V_2c_128Mb_2c_v1_0-371013.pdf
- [5] Abracon, *ASE series datasheet*, pin layout and supply/enable functions. <https://abracon.com/Oscillators/ASEseries.pdf>
- [6] Samtec, QSH suggested footprint drawing. Exact mating height, companion connector and assembly envelope remain open. <https://suddendocs.samtec.com/prints/qsh-xxx-01-x-d-xx-footprint.pdf>
- [7] Local core design: component_manifest_resolved.json, FPGA100T_Minimal.net; actual revised board: FPGA100T_33x36_Placement.kicad_pcb. These establish saved assignments, not measured hardware behaviour.
- [8] Local interface proposal: Mezzanine_Contact_Proposal.csv; team correction from 113 to 117 includes four additional SPI clocks. Earlier ASIC-slide reviews and the full open-uncertainty register remain in the dated review folder and on the project review site.